

Gonzalo Barrera Borla

Selected Experience

Industry (est. 2013)

- 2024 – **Scientist, Anti-Fraud Team, Muun Wallet**, Buenos Aires
Our team is responsible for curbing fraud-related operating expenses without damaging the UX for our BTC wallet's hundreds of thousands of legitimate users. As a scientist, I am in charge of detecting common fraud patterns, devising product strategies to mitigate them, and deploying them to production in collaboration with engineers.
- 2023 – **Freelance ML/AI Consulting, Borlandux**, Remote
I started my own one-man consulting shop, delivering end-to-end ML solutions for companies facing tough statistical challenges without a dedicated in-house team. Projects include a profit-optimization strategy for the largest horseracing betting syndicate in LATAM, and a labor time prediction model for a regional frozen goods logistics company in the USA.
- 2021 – 22 **Engineer Manager, Data Science, Jampp**, Remote
I defined development priorities for my team alongside Tech and Product managers. I coordinated the migration from in-house GLMs estimators to LightGBM. We professionalized the team, by implementing agile methodologies, documenting internal libraries and setting reproducibility standards for our analysis. I wrote the first career ladder, updated the technical exams and grew the team from 7 to 12 people.
- 2019 – 20 **Senior Data Scientist, Jampp**, Buenos Aires
I assisted in the development of the income optimization engine for our Real Time Bidding (RTB) service. We presented the results in the [2019 KDD](#). Our team was responsible for the system predicting ads' conversion rates (win rate, CTR, CVR). We trained cheaply on several billions observations with an in-house, online SGD algo, and processed 1M+ auctions per seconds, using strongly regularised GLMs.
- 2016 – 19 **Expert Data Analysis Advisor, Jefatura de Gabinete de Ministros**, BA
Our team implemented an end-to-end business intelligence system to integrate data from key systems of the National Public Administration (e.g. Budget, Payroll, Public Policies). I worked in defining the data model definition and developing ETLs, apart from advising the Justice and Economy ministries on how to define their OKRs within the system.

Academia (est. 2006)

- 2023 – 25 **Teching Assistant, 1st Class, Instituto del Cálculo, Facultad de Cs. Exactas y Naturales (FCEyN), UBA**, Buenos Aires
As TA for the Statistics department, I've taught "Intro to Statistics and Data Science" (4th year) and "Data Laboratory" (1st year).
- 2021 H2 **External Professor, FCEyN, Fac. de Ingeniería, UBA**, Buenos Aires, Virtual
I taught "Machine Learning in Graphs", with Prof. Martín Elías Costa, Ph.D. Video lectures available at [Youtube](#), exercise guides and didactic materials at [GitHub](#).
- 2012 – 15 **Teaching Assistant, 2nd Class (Statistics II), Fac. de Cs. Económicas, UBA**
- 2012 – 17 **Mathematical Olympiad Coach**, Buenos Aires
- 2006 – 16 **Private Tutor (Math, Physics, Chemistry)**, Buenos Aires

Academic Record

2017 – 21 **MSc in Mathematical Statistics**, *Instituto del Cálculo, FCEyN, UBA*
GPA 10.0

Thesis **Fermat Distance in Kernel Density Estimators**, in progress

I propose to train a KDE classifier with a data-learned distance - Fermat's Distance - instead of the customary euclidean distance in order to learn not only the sample density, but also its geometry (i.e. its supporting Riemannian manifold). I evaluate the classifier against related algorithms such as Naive Bayes, and state-of-the art techniques such as GBTs and NNs. Publicly available in GitHub [\[link\]](#). Director: Pablo Groisman, Ph.D.

2009 – 14 **Licenciatura (~MSc) en Economía**, *Facultad de Cs. Económicas, UBA*
GPA 8.94. I graduated *magna cum laude* as class valedictorian.

Thesis **The limits of electoral predictability**

Using open data from 2013 legislative elections, I first did a simulation study to estimate the minimum possible error in predicting the election's final result. In light of the results, I then assessed the credibility of several public opinion polls published right before the voting day. Publicly available in GitHub [\[link\]](#). Director: Silvia Vietri, Ph.D. Grade: 10.0.

Languages

- Spanish - Native
 - French - Advanced
 - English - Bilingual *CPE (CEFR C2), 2008.*
 - Portuguese - Advanced
- Grade: A*

Computer Skills

Languages Python (incl. scientific stack), SQL, R, Unix Scripting, Ruby
Data RDBMS (PostgreSQL), Tableau, PowerBI, Plotly, Prefect, Airflow, Pentaho
DevOps Major cloud providers (AWS, Azure & GCP), containerization, basic CI/CD
Other Working knowledge of HTML/CSS, Ruby, MATLAB, $\text{\LaTeX}$, Typst

Other things I've Done and Like

- In my spare time, I investigate the use of agentic AI tooling to improve my productivity, by building development workflows to leverage my secondary skills through PR-style codebase changes and MCP-based actions.
- At [PyData 2019](#) I presented “Optimal Bidding: a dual approach” to the local community, premired earlier in the year by a teammate at [AdKDD 2019](#).
- [fcen-amateur](#) is an open GitHub organization in which I share solved exams, lectures & study guides from every course I've taken or given at FCEyN, fostering programming practices in the academic community.
- [pydatajson](#), a library for managing [CKAN](#) metadata, is part of the Inter-American Development Bank “[Code for Development](#)” FOSS program.
- I really enjoyed [Machine Learning](#), from Stanford at Coursera, teaching to code fundamental machine learning algorithms from scratch.
- Mathematical and Chemistry olympiads were my favorite K-12 activity. I got international awards in the former, and national ones in the latter.